

Apache_SeaTunnel_Guide

Apache SeaTunnel — Installation & Verification Guide

Version 2.3.13 • Big Data Tools & Applications

 **Path:** Wherever you see /home/ubuntu or /home/shrivardhan in commands, configs, or screenshots — replace the username with YOUR Linux username (e.g. /home/ravi/...).

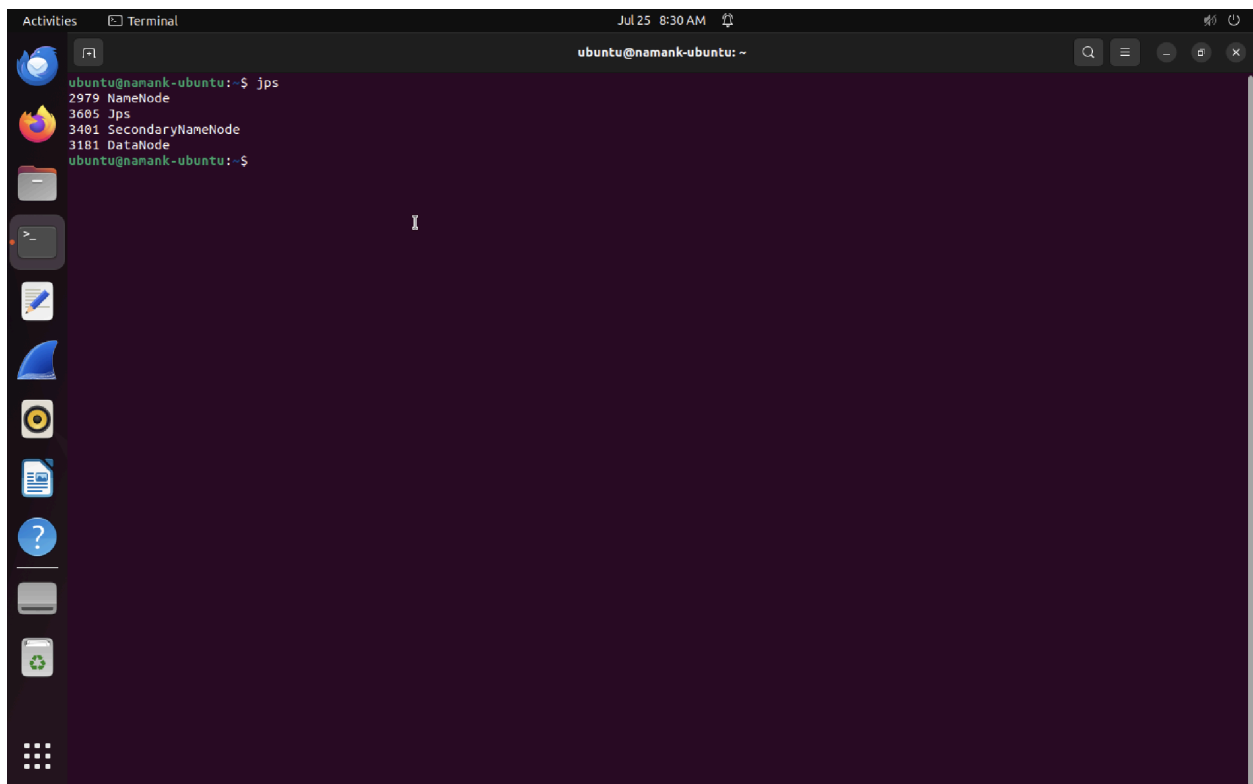
1. Prerequisites

Component	Requirement
OS	Ubuntu 22.04+ (VM), 4GB+ RAM, 2+ CPUs
Java	OpenJDK 11 (mandatory — JAVA_HOME must be set)
Hadoop	3.3.6, pseudo-distributed mode, running

2. Verify HDFS & Java

```
jps
```

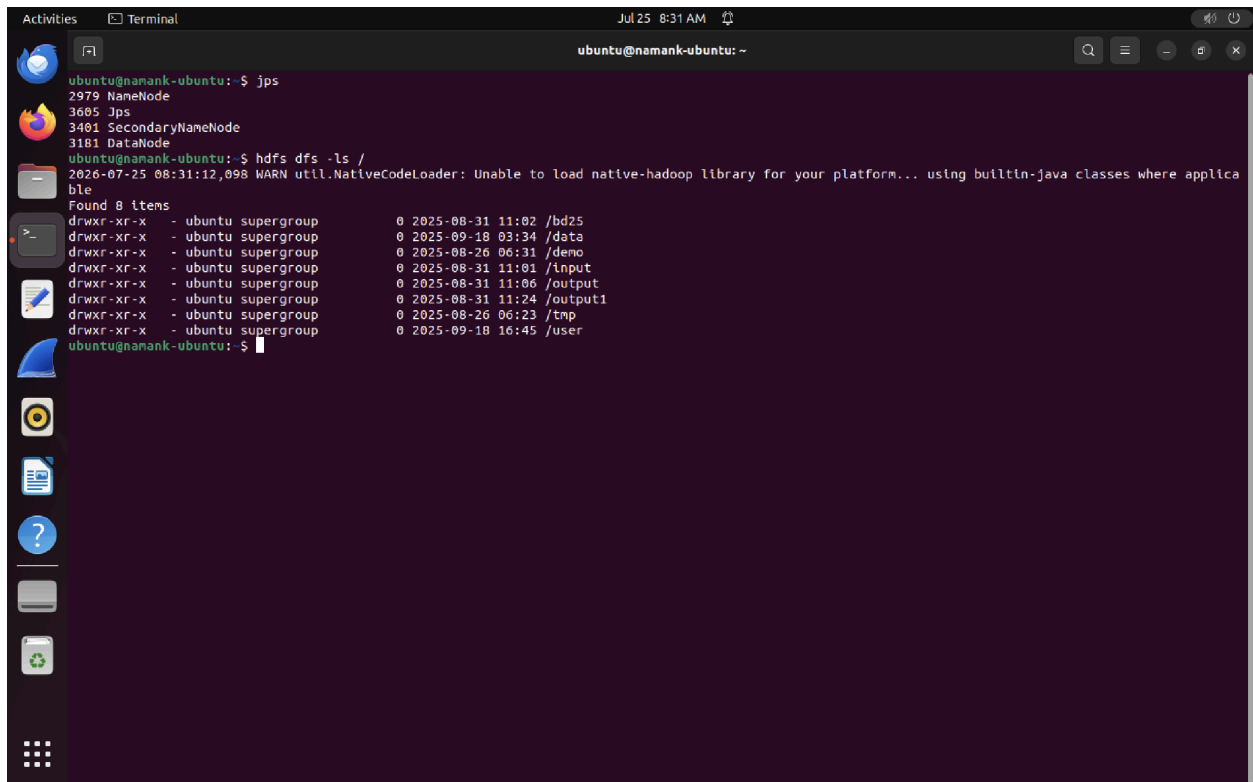
Must show: NameNode, DataNode, SecondaryNameNode.

A screenshot of a Linux terminal window. The window title is "Terminal" and it shows the date and time as "Jul 25 8:30 AM". The prompt is "ubuntu@namank-ubuntu: ~". The user has entered the command "jps". The output shows four processes: "2979 NameNode", "3605 Jps", "3401 SecondaryNameNode", and "3181 DataNode". The prompt returns to "ubuntu@namank-ubuntu: \$".

```
ubuntu@namank-ubuntu:~$ jps
2979 NameNode
3605 Jps
3401 SecondaryNameNode
3181 DataNode
ubuntu@namank-ubuntu:~$
```

jps output — Hadoop daemons running

```
hdfs dfs -ls /
```

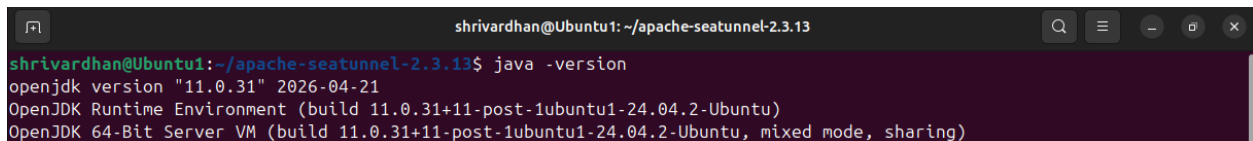
A terminal window on an Ubuntu system. The user runs 'jps' and 'hdfs dfs -ls /'. The 'jps' command shows three processes: NameNode (PID 2979), Jps (PID 3605), and SecondaryNameNode (PID 3401). The 'hdfs dfs -ls /' command shows a warning about the native-hadoop library and then lists the root directory contents. The output shows eight items: bd25, data, demo, input, output, output1, tmp, and user, each with permissions, owner, group, size, and modification time.

```
ubuntu@namank-ubuntu:~$ jps
2979 NameNode
3605 Jps
3401 SecondaryNameNode
ubuntu@namank-ubuntu:~$ hdfs dfs -ls /
2026-07-25 08:31:12,098 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 8 items
drwxr-xr-x - ubuntu supergroup          0 2025-08-31 11:02 /bd25
drwxr-xr-x - ubuntu supergroup          0 2025-09-18 03:34 /data
drwxr-xr-x - ubuntu supergroup          0 2025-08-26 06:31 /demo
drwxr-xr-x - ubuntu supergroup          0 2025-08-31 11:01 /input
drwxr-xr-x - ubuntu supergroup          0 2025-08-31 11:06 /output
drwxr-xr-x - ubuntu supergroup          0 2025-08-31 11:24 /output1
drwxr-xr-x - ubuntu supergroup          0 2025-08-26 06:23 /tmp
drwxr-xr-x - ubuntu supergroup          0 2025-09-18 16:45 /user
ubuntu@namank-ubuntu:~$
```

HDFS root listing

```
java -version
```

Must show Java 11.

A terminal window showing the output of the 'java -version' command. The output indicates that OpenJDK version 11.0.31 is installed, built on 2026-04-21, and is a 64-bit server VM.

```
shrivardhan@Ubuntu1: ~/apache-seatunnel-2.3.13
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-24.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-24.04.2-Ubuntu, mixed mode, sharing)
```

java -version output

⚠️: If any Hadoop daemon is missing or HDFS is not accessible, fix Hadoop first.

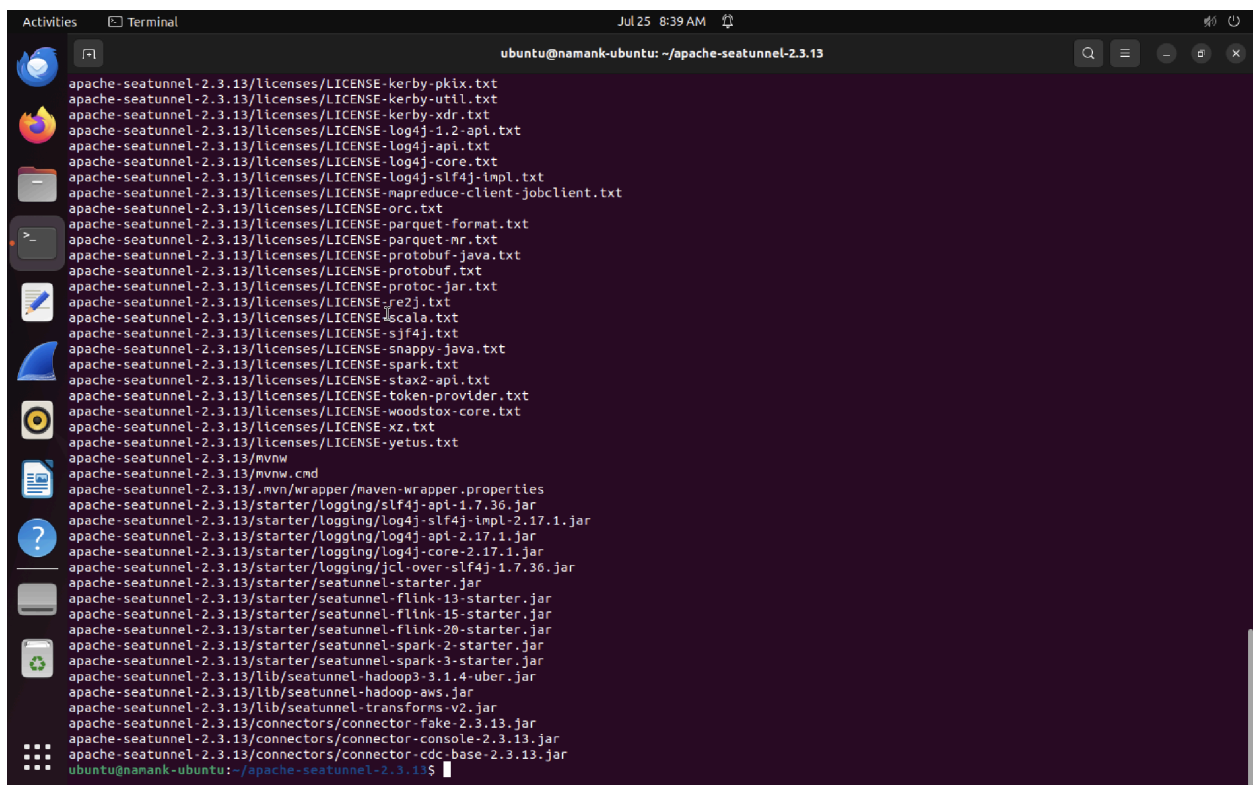
3. Download & Extract SeaTunnel

```
cd ~
```

```
wget https://archive.apache.org/dist/seatunnel/2.3.13/apache-seatunnel-2.3.13-bin.tar.gz
```

```
tar -xzf apache-seatunnel-2.3.13-bin.tar.gz
```

```
cd apache-seatunnel-2.3.13
```



SeaTunnel directory after extraction

4. Configure & Install Plugins

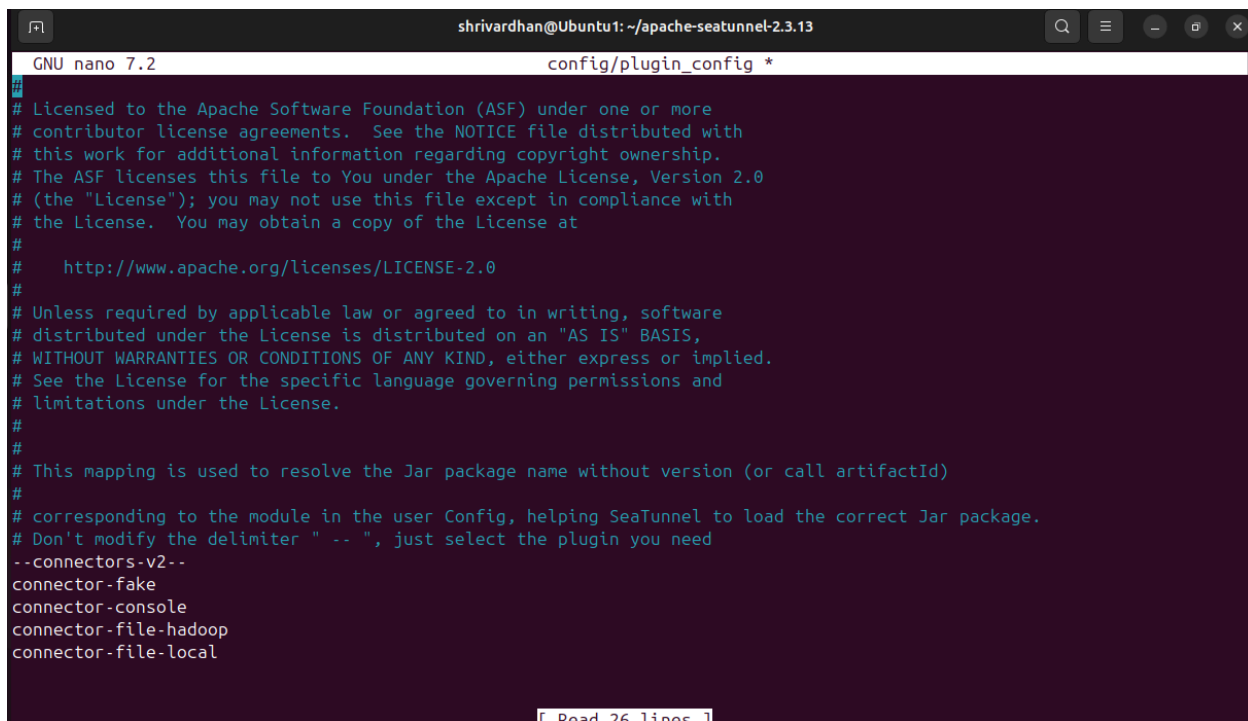
Open the plugin config and set it to include all four connectors needed for every task in this guide:

```
nano config/plugin_config
```

Contents:

(Note:Initially there will be a lot of connectors which we do not need. Replace them with only these 4 connectors)

```
--connectors-v2--  
connector-fake  
connector-console  
connector-file-hadoop  
connector-file-local
```



The screenshot shows a terminal window titled "shrivardhan@Ubuntu1: ~/apache-seatunnel-2.3.13". The window displays the contents of the file "config/plugin_config *" using the GNU nano 7.2 editor. The file contains a license notice for the Apache Software Foundation (ASF) and a list of connectors to be installed. The connectors listed are: connector-fake, connector-console, connector-file-hadoop, and connector-file-local. The terminal window also shows a status bar at the bottom indicating "Read 26 lines".

```
GNU nano 7.2 config/plugin_config *  
# Licensed to the Apache Software Foundation (ASF) under one or more  
# contributor license agreements. See the NOTICE file distributed with  
# this work for additional information regarding copyright ownership.  
# The ASF licenses this file to You under the Apache License, Version 2.0  
# (the "License"); you may not use this file except in compliance with  
# the License. You may obtain a copy of the License at  
#  
# http://www.apache.org/licenses/LICENSE-2.0  
#  
# Unless required by applicable law or agreed to in writing, software  
# distributed under the license is distributed on an "AS IS" BASIS,  
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.  
# See the License for the specific language governing permissions and  
# limitations under the License.  
#  
# This mapping is used to resolve the Jar package name without version (or call artifactId)  
#  
# corresponding to the module in the user Config, helping SeaTunnel to load the correct Jar package.  
# Don't modify the delimiter " -- ", just select the plugin you need  
--connectors-v2--  
connector-fake  
connector-console  
connector-file-hadoop  
connector-file-local  
[ Read 26 lines ]
```

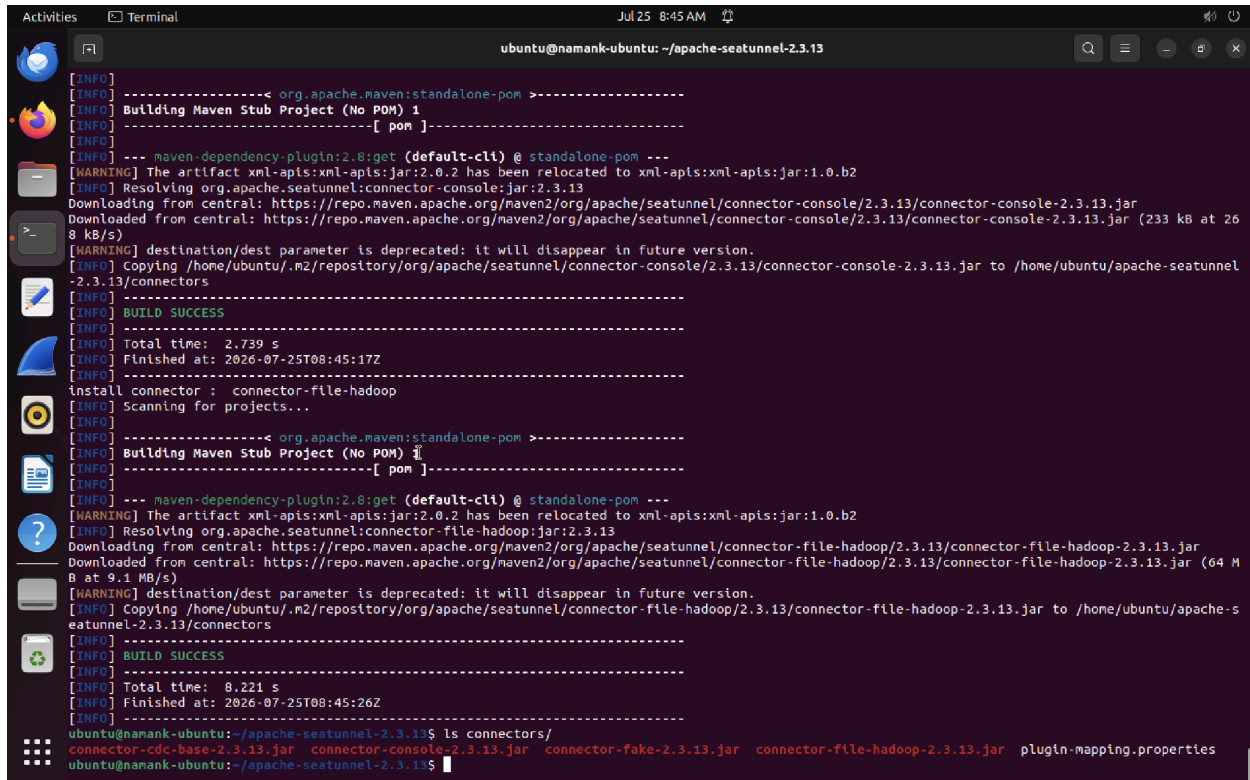
plugin_config with all 4 connectors

Install them:

```
sh bin/install-plugin.sh
```

Verify JARs are downloaded:

```
ls connectors/
```



```
ubuntu@namank-ubuntu: ~/apache-seatunnel-2.3.13
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] --- maven-dependency-plugin:2.8:get (default-cli) @ standalone-pom ---
[WARNING] The artifact xml-apis:xml-apis:jar:2.0.2 has been relocated to xml-apis:xml-apis:jar:1.0.b2
[INFO] Resolving org.apache.seatunnel:connector-console:jar:2.3.13
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/seatunnel/connector-console/2.3.13/connector-console-2.3.13.jar (233 kB at 268 kB/s)
[WARNING] destination/dest parameter is deprecated: it will disappear in future version.
[INFO] Copying /home/ubuntu/.m2/repository/org/apache/seatunnel/connector-console/2.3.13/connector-console-2.3.13.jar to /home/ubuntu/apache-seatunnel-2.3.13/connectors
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.739 s
[INFO] Finished at: 2026-07-25T08:45:17Z
[INFO] -----
Install connector : connector-file-hadoop
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] --- maven-dependency-plugin:2.8:get (default-cli) @ standalone-pom ---
[WARNING] The artifact xml-apis:xml-apis:jar:2.0.2 has been relocated to xml-apis:xml-apis:jar:1.0.b2
[INFO] Resolving org.apache.seatunnel:connector-file-hadoop:jar:2.3.13
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/seatunnel/connector-file-hadoop/2.3.13/connector-file-hadoop-2.3.13.jar (64 MB at 9.1 MB/s)
[WARNING] destination/dest parameter is deprecated: it will disappear in future version.
[INFO] Copying /home/ubuntu/.m2/repository/org/apache/seatunnel/connector-file-hadoop/2.3.13/connector-file-hadoop-2.3.13.jar to /home/ubuntu/apache-seatunnel-2.3.13/connectors
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.221 s
[INFO] Finished at: 2026-07-25T08:45:26Z
[INFO] -----
ubuntu@namank-ubuntu:~/apache-seatunnel-2.3.13$ ls connectors/
connector-cdc-base-2.3.13.jar  connector-console-2.3.13.jar  connector-fake-2.3.13.jar  connector-file-hadoop-2.3.13.jar  plugin-mapping.properties
ubuntu@namank-ubuntu:~/apache-seatunnel-2.3.13$
```

Plugin installation output + connectors listing

Task 1 — Local CSV → Console (FieldMapper)

Pipeline: students.csv → LocalFile → FieldMapper → Console

Transformation

- student_name → name (rename)
- score → marks (rename)
- age → dropped
- student_id, department → kept as-is

1.1 Create dataset

```
mkdir -p ~/seatunnel-verification
```

```
nano ~/seatunnel-verification/students.csv
```

Paste:

```
student_id,student_name,age,score,department
```

```
101,Aarav,20,87,CSE
```

```
102,Diya,21,92,ECE
```

```
103,Rohan,20,76,ISE
```

```
104,Ananya,22,95,CSE
```

```
105,Vikram,21,68,ME
```

```
106,Meera,20,89,ECE
```

```
107,Arjun,22,73,ISE
```

```
108,Ishita,21,91,CSE
```

```
109,Karan,20,64,ME
```

```
110,Sneha,22,88,ECE
```

```
111,Rahul,21,79,ISE
```

```
112,Priya,20,94,CSE
```

```
113,Aditya,22,71,ME
114,Neha,21,86,ECE
115,Manish,20,77,ISE
116,Kavya,22,96,CSE
117,Nikhil,21,69,ME
118,Riya,20,90,ECE
119,Varun,22,74,ISE
120,Tanvi,21,93,CSE
```

Verify (expect 21 lines):

```
wc -l ~/seatunnel-verification/students.csv
```

```
shrivardhan@Ubuntu1: ~/apache-seatunnel-2.3.13
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ mkdir -p ~/seatunnel-verification
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ nano ~/seatunnel-verification/students.csv
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ wc -l ~/seatunnel-verification/students.csv
21 /home/shrivardhan/seatunnel-verification/students.csv
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$
```

students.csv created — 21 lines

1.2 Config file

```
nano ~/seatunnel-verification/task1_local_console.conf
```

Paste:

```
env {
  parallelism = 1
  job.mode = "BATCH"
}

source {
  LocalFile {
```

```

    path = "/home/<YOUR_USERNAME>/seatunnel-verification/stud
ents.csv"
    file_format_type = "csv"
    skip_header_row_number = 1

    schema = {
      fields {
        student_id = int
        student_name = string
        age = int
        score = int
        department = string
      }
    }
  }
}

transform {
  FieldMapper {
    field_mapper = {
      student_id = student_id
      student_name = name
      score = marks
      department = department
    }
  }
}

sink {
  Console {}
}

```

: Replace <YOUR_USERNAME> with your Linux username (e.g. /home/ravi/...).

1.3 Run

```
cd ~/apache-seatunnel-2.3.13
```

```
./bin/seatunnel.sh --config ~/seatunnel-verification/task1_local_console.conf -m local
```

Expected: 20 read, 20 written, 0 failed.

```
*****
                        Job Statistic Information
*****
Start Time           : 2026-08-13 05:17:52
End Time             : 2026-08-13 05:17:54
Total Time(s)        : 2
Total Read Count      : 20
Total Write Count     : 20
Total Failed Count    : 0
*****
```

Task 1 result — 20 read, 20 written, 0 failed

Task 2 — Local CSV → HDFS (SQL Transform)

Pipeline: employees.csv → LocalFile → SQL → HdfsFile → HDFS

Transformation

- "Eng" → "Engineering"
- "HR" → "Human Resources"
- "Sales" → unchanged
- salary increased by 10% (salary * 11 / 10)

2.1 Create dataset

```
nano ~/seatunnel-verification/employees.csv
```

Paste:

```
employee_id,name,department,city,salary
```

```
201,Raj,Eng,Bengaluru,72000
```

```
202,Simran,Sales,Mumbai,68000
```

```
203,Dev,HR,Delhi,61000
```

```
204,Pooja,Eng,Pune,75000
```

```
205,Akash,Sales,Chennai,64000
```

```
206,Nandini,HR,Hyderabad,62000
```

```
207,Manav,Eng,Bengaluru,81000
```

```
208,Aditi,Sales,Mumbai,71000
```

```
209,Kunal,HR,Delhi,59000
```

```
210,Shreya,Eng,Pune,78000
```

```
211,Rohit,Sales,Chennai,67000
```

```
212,Ira,HR,Hyderabad,63000
```

```
213,Sameer,Eng,Bengaluru,83000
```

```
214,Anjali,Sales,Mumbai,69000
```

215,Varun,HR,Delhi,60000

216,Deepa,Eng,Pune,77000

217,Arnav,Sales,Chennai,66000

218,Myra,HR,Hyderabad,65000

219,Yash,Eng,Bengaluru,79000

220,Kiara,Sales,Mumbai,73000

Verify (expect 21 lines):

```
wc -l ~/seatunnel-verification/employees.csv
```

```
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ nano ~/seatunnel-verification/employees.csv
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ wc -l ~/seatunnel-verification/employees.csv
21 /home/shrivardhan/seatunnel-verification/employees.csv
```

employees.csv — 21 lines

2.2 Config file

```
nano ~/seatunnel-verification/task2_local_hdfs.conf
```

Paste:

```
env {
  parallelism = 1
  job.mode = "BATCH"
}
```

```

source {
  LocalFile {
    path = "/home/<YOUR_USERNAME>/seatunnel-verification/employees.csv"
    file_format_type = "csv"
    skip_header_row_number = 1

    schema = {
      fields {
        employee_id = int
        name = string
        department = string
        city = string
        salary = int
      }
    }

    plugin_output = "employees"
  }
}

transform {
  Sql {
    plugin_input = "employees"
    plugin_output = "employees_transformed"

    query = "select employee_id, name, case when department = 'Eng' then 'Engineering' when department = 'HR' then 'Human Resources' else department end as department, city, salary * 11 / 10 as salary from dual"
  }
}

sink {
  HdfsFile {
    plugin_input = "employees_transformed"
  }
}

```

```

    fs.defaultFS = "hdfs://127.0.0.1:9000"
    path = "/seatunnel-verification/employees"
    file_format_type = "csv"
    field_delimiter = ","
    write_mode = "overwrite"
}
}

```

📌: Replace <YOUR_USERNAME> with your Linux username. The SQL query goes on a single line in the actual file.

2.3 Run

```
cd ~/apache-seatunnel-2.3.13
```

```
./bin/seatunnel.sh --config ~/seatunnel-verification/task2_local_hdfs.conf -m local
```

Expected: 20 read, 20 written, 0 failed.

```

*****
                Job Statistic Information
*****
Start Time           : 2026-08-13 05:28:59
End Time             : 2026-08-13 05:29:03
Total Time(s)        : 4
Total Read Count      : 20
Total Write Count     : 20
Total Failed Count    : 0
*****

```

Task 2 result — 20 read, 20 written, 0 failed

2.4 Verify HDFS output

```
hdfs dfs -ls /seatunnel-verification/employees
```

```
hdfs dfs -cat /seatunnel-verification/employees/*
```

Eng → Engineering, HR → Human Resources, salaries increased by 10%.

```
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ hdfs dfs -ls /seatunnel-verification/employees
2026-08-13 05:29:29,381 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
Found 1 items
-rw-r--r-- 3 shrivardhan supergroup 754 2026-08-13 05:29 /seatunnel-verification/employees/T_1140155018398138369_1f263702fc_0_1_0.csv
shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$ hdfs dfs -cat /seatunnel-verification/employees/*
2026-08-13 05:29:42,282 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using built-in java classes where applicable
201,Raj,Engineering,Bengaluru,79200.0
202,Simran,Sales,Mumbai,74800.0
203,Dev,Human Resources,Delhi,67100.0
204,Pooja,Engineering,Pune,82500.0
205,Akash,Sales,Chennai,70400.0
206,Nandini,Human Resources,Hyderabad,68200.0
207,Manav,Engineering,Bengaluru,89100.0
208,Aditi,Sales,Mumbai,78100.0
209,Kunal,Human Resources,Delhi,64900.000000000001
210,Shreya,Engineering,Pune,85800.0
211,Rohit,Sales,Chennai,73700.0
212,Ira,Human Resources,Hyderabad,69300.0
213,Sameer,Engineering,Bengaluru,91300.000000000001
214,Anjali,Sales,Mumbai,75900.0
215,Varun,Human Resources,Delhi,66000.0
216,Deepa,Engineering,Pune,84700.0
217,Arnav,Sales,Chennai,72600.0
218,Myra,Human Resources,Hyderabad,71500.0
219,Yash,Engineering,Bengaluru,86900.0
220,Kiara,Sales,Mumbai,80300.0shrivardhan@Ubuntu1:~/apache-seatunnel-2.3.13$
```

HDFS output — transformed records

What SeaTunnel Actually Does

Given a dataset (local or on HDFS), SeaTunnel does the following:

- Reads the file in splits, parsing raw text per the declared format (CSV, JSON, etc.).
- Structures each row into typed, named columns using the schema block — turns raw text into a proper table.
- Transforms the data if a transform block is present — filtering, renaming, casting, aggregating.

- Writes the result to the configured sink in a different format (Parquet, CSV to HDFS, Console, etc.).

This is essentially the same CSV-to-columnar-format conversion that underlies most real data lake ingestion pipelines — configured declaratively, no code needed.

Troubleshooting

Connector Not Found

Plugin not installed. Fix:

```
nano config/plugin_config # add missing connector  
  
sh bin/install-plugin.sh  
  
ls connectors/
```

HdfsFile Not Recognized

connector-file-hadoop JAR missing. Reinstall plugins and check:

```
ls connectors/ | grep hadoop
```

Cannot Connect to HDFS

Usually a Hadoop problem. Check Hadoop first:

```
jps  
  
hdfs dfs -ls /
```

If those work, verify `fs.defaultFS` in your config matches your Hadoop setup (usually `hdfs://localhost:9000`).

"Unable to load native-hadoop library" warning

Safe to ignore. Hadoop uses pure-Java fallback — everything still works.

Wrong row counts

Common causes:

- Schema fields don't match actual CSV columns (order or types)
- `skip_header_row_number` wrong (0 vs 1)
- Different delimiter (tabs/semicolons instead of commas)

Go back and inspect the raw data: `head`, `wc -l`, `file`.